

Frames and matrix designs

Vanishing injectivity probability at $4d$ minus 5 complex phase measurements

Difficulty: challenging **Importance:** interesting to the community**Status:** Solved

Rating rationale: The asymptotic typical-injectivity question needs quantitative algebraic and probabilistic control beyond positive failure probability; it informs phase-retrieval measurement design.

Original statement

For every integer $d \geq 2$, draw $A_d \in \mathbb{C}^{(4d-5) \times d}$ with independent standard complex Gaussian entries: the real and imaginary parts are independent $N(0, 1/2)$. Let p_d be the probability that

$$|A_d x| = |A_d y| \implies y = e^{i\theta} x \text{ for some } \theta \in \mathbb{R}$$

holds for all $x, y \in \mathbb{C}^d$, where absolute value is componentwise. Equivalently, p_d is the probability that the induced phase-retrieval map on vectors modulo global phase is injective.

Is

$$\lim_{d \rightarrow \infty} p_d = 0?$$

This is part (b) of Vinzant's conjecture as restated in Randomstrasse 101. It distinguishes injective exceptional measurement systems from typical matrices at a row count just below $4d-4$. Injectivity is the basic identifiability requirement before conditioning and stable numerical inversion can be addressed.

Resolution — independently audited 8 October 2026

Zhangsong Li's [13-page manuscript](#), Theorem 1.4, proves the stronger bound

$$p_d \leq C/d \quad (d \geq 2)$$

for an absolute constant $C > 0$. This implies the displayed limit. The argument plants a two-dimensional ambiguity, compares its law in L^2 to a covariance-matched Gaussian law, and uses a quantitative implicit-function argument to turn the planted approximate ambiguity into an exact one.

An [independent AI-agent proof audit](#) checked the manuscript's planted and reference laws, the cone and overlap estimates, the small-ball and Newton arguments, and the final probability reduction. No proof gap was found. This supports **Solved** under the catalog's informal-audit rule; it is not external human peer review or *Lean verified*. The original target, permanent identifier, and canonical path are unchanged.

Local Lean development — 27 September 2026

The [Lean solution](#) now proves the bound $p_d \leq C/d$ for every $d \geq 2$ and the original limit $p_d \rightarrow 0$, using the original Gaussian law and all-signals injectivity predicate. The recorded local build and transitive axiom audit pass, using only standard Lean axioms and no proof placeholders in the solution. See the [final assembly](#), [source guide](#), and [local verification record](#). An independent statement review and isolated Linux Comparator/kernel verification remain outstanding for *Lean verified*; the 8 October promotion rests on the manuscript proof audit above.

References

1. A. S. Bandeira et al., *Randomstrasse 101: Open Problems of 2025*, arXiv:2603.29571 (2026), Conjecture 19(b). [Paper](#).
2. C. Vinzant, *A small frame and a certificate of its injectivity*, SAMPTA 2015, arXiv:1502.04656. The explicit 11-vector frame in $\mathbb{C}^4$ demonstrates why the below- $4d-4$ regime cannot simply be dismissed. [Paper](#).
3. Z. Li, *Resolution of Vinzant's Conjecture on Phase Retrieval Injectivity* (13 September 2026), Theorem 1.4. The manuscript proves $p_d \leq C/d$ for every $d \geq 2$; the audited copy is pinned by SHA-256 in the [review](#). [Manuscript](#).
4. Z. Li, *On Injectivity of Phase Retrieval*, arXiv:2606.17922 (2026). This earlier version established only $p_d < 1$. [Paper](#).

Historical status check — 2026-09-10

Searched “Vinzant conjecture injectivity probability limit”, “phase retrieval 4M-5 2026”, and checked the latest June 2026 paper abstract/version. Its positive probability of noninjectivity does not supply an asymptotic lower bound tending to one. The already proved assertion $p_d < 1$ is deliberately excluded from this problem.

Audit update (2026-09-10): Rechecked Li’s June 2026 abstract and searched for the asymptotic part of Vinzant’s conjecture. The reported strict inequality $p_d < 1$ does not force $p_d \rightarrow 0$; this entry contained only the latter target. This was a bounded literature check, not a proof that no solution existed at that time.